# A First Course in Abstract Algebra

## Section 35: Series of Groups

Detailed worked solutions

Mathjock

**Text and numbering.** John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

**How to read the solutions.** Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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# 35 Series of Groups

Throughout this section $D_4 = \langle r, s\rangle$ denotes the order-8 square-symmetry group, with $r^4 = s^2 = e$ and $srs = r^{-1}$. We follow the textbook's definitions of principal series and solvability; qualifications are stated where they affect an answer.

### Exercise 35.1: Refining the $10\mathbb{Z}$ and $25\mathbb{Z}$ series

Insert subgroups to obtain

$$\{0\} < 50\mathbb{Z} < 10\mathbb{Z} < 5\mathbb{Z} < \mathbb{Z},$$
$$\{0\} < 50\mathbb{Z} < 25\mathbb{Z} < 5\mathbb{Z} < \mathbb{Z}.$$

The first factor in each is $50\mathbb{Z} \cong \mathbb{Z}$. The remaining factors are, respectively, $\mathbb{Z}_5, \mathbb{Z}_2, \mathbb{Z}_5$ and $\mathbb{Z}_2, \mathbb{Z}_5, \mathbb{Z}_5$. Thus the factors match after reordering. Here $a\mathbb{Z}/b\mathbb{Z} \cong \mathbb{Z}_{b/a}$ when $a \mid b$; remember that a larger integer labels a smaller subgroup.

### Exercise 35.2: Refining the $60\mathbb{Z}$ and $245\mathbb{Z}$ series

A common initial nonzero subgroup is $2940\mathbb{Z}$, since $\operatorname{lcm}(60, 245) = 2940$. Use

$$\begin{aligned}\{0\} < 2940\mathbb{Z} < 420\mathbb{Z} < 60\mathbb{Z} < 20\mathbb{Z}\\ < 10\mathbb{Z} < 5\mathbb{Z} < \mathbb{Z},\\ \{0\} < 2940\mathbb{Z} < 1470\mathbb{Z} < 735\mathbb{Z} < 245\mathbb{Z}\\ < 49\mathbb{Z} < 7\mathbb{Z} < \mathbb{Z}.\end{aligned}$$

Every original subgroup remains. In the first series the factors after $2940\mathbb{Z} \cong \mathbb{Z}$ have orders $7, 7, 3, 2, 2, 5$; in the second they have orders $2, 2, 3, 5, 7, 7$. Each factor is cyclic, so this is an isomorphism of series.

### Exercise 35.3: Refinements in $\mathbb{Z}_{24}$

All subgroups here are cyclic and normal. The refinements

$$\{0\} < \langle 12\rangle < \langle 6\rangle < \langle 3\rangle < \mathbb{Z}_{24},$$
$$\{0\} < \langle 8\rangle < \langle 4\rangle < \langle 2\rangle < \mathbb{Z}_{24}$$

have factor orders $2, 2, 2, 3$ and $3, 2, 2, 2$. They retain respectively $\langle 3\rangle$ and $\langle 8\rangle$. Each factor is cyclic of the indicated prime order, giving the required matching.

### Exercise 35.4: Refinements in $\mathbb{Z}_{72}$

Use

$$\{0\} < \langle 36\rangle < \langle 18\rangle < \langle 9\rangle < \langle 3\rangle < \mathbb{Z}_{72},$$
$$\{0\} < \langle 24\rangle < \langle 12\rangle < \langle 6\rangle < \langle 3\rangle < \mathbb{Z}_{72}.$$

The orders of these subgroups are respectively $1, 2, 4, 8, 24, 72$ and $1, 3, 6, 12, 24, 72$. Consecutive order ratios give cyclic factors of orders $2, 2, 2, 3, 3$ and $3, 2, 2, 2, 3$. These agree up to order and all original terms remain.

### Exercise 35.5: Refining two series of $\mathbb{Z}^2$

Put $L = 60\mathbb{Z} \times 80\mathbb{Z}$ and $0 = \{(0,0)\}$. The refinements are

$$\begin{aligned} 0 < L < 60\mathbb{Z} \times 20\mathbb{Z} < 60\mathbb{Z} \times \mathbb{Z} \\ < 10\mathbb{Z} \times \mathbb{Z} < \mathbb{Z}^2, \\ 0 < L < 10\mathbb{Z} \times 80\mathbb{Z} < \mathbb{Z} \times 80\mathbb{Z} \\ < \mathbb{Z} \times 20\mathbb{Z} < \mathbb{Z}^2. \end{aligned}$$

The initial factor is $L \cong \mathbb{Z}^2$ in both. The other factors are $\mathbb{Z}_4, \mathbb{Z}_{20}, \mathbb{Z}_6, \mathbb{Z}_{10}$ in the first and $\mathbb{Z}_6, \mathbb{Z}_{10}, \mathbb{Z}_4, \mathbb{Z}_{20}$ in the second. This follows by quotienting the coordinate that changes at each step; the unchanged coordinate cancels.

### Exercise 35.6: All composition series of $\mathbb{Z}_{60}$

**Plan.** A cyclic group has exactly one subgroup $U_d$ of each order $d \mid 60$, namely $U_d = \langle 60/d \rangle$. A cyclic factor is simple exactly when its order is prime. Consequently *all* the series are

$$U_1 < U_{p_1} < U_{p_1 p_2} < U_{p_1 p_2 p_3} < U_{60},$$

where $(p_1, p_2, p_3, p_4)$ runs over the distinct permutations of $(2, 2, 3, 5)$. This is an explicit parametrization with no duplication: the subgroup orders recover the sequence of primes. There are $4!/2! = 12$ such series. All have factors $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_5$, in some order, so all are isomorphic.

### Exercise 35.7: All composition series of $\mathbb{Z}_{48}$

Let $U_d = \langle 48/d \rangle$ be the unique subgroup of order $d$. The possible successive subgroup orders are

$$\begin{gathered} 1, 3, 6, 12, 24, 48 \\ 1, 2, 6, 12, 24, 48 \\ 1, 2, 4, 12, 24, 48 \\ 1, 2, 4, 8, 24, 48 \\ 1, 2, 4, 8, 16, 48. \end{gathered}$$

Replacing every order $d$ by $U_d$ lists all five series. Indeed each step must multiply the order by a prime, and $48 = 2^4 \cdot 3$, so the only choice is the position of the factor 3 among five steps. Each series has four factors $\mathbb{Z}_2$ and one $\mathbb{Z}_3$.

### Exercise 35.8: Composition series of $\mathbb{Z}_5^2$

Regard $\mathbb{Z}_5^2$ as a two-dimensional vector space over $\mathbb{Z}_5$. Its order-5 subgroups are exactly the six lines

$$L_t = \langle (1, t) \rangle \ (t = 0, 1, 2, 3, 4), \qquad L_\infty = \langle (0, 1) \rangle.$$

They exhaust the possibilities: a nonzero vector with nonzero first coordinate scales uniquely to $(1, t)$; otherwise it scales to $(0, 1)$. The six composition series are $0 < L_t < \mathbb{Z}_5^2$, including $t = \infty$. Both factors have order 5. No other composition series is possible, since simple factors of an abelian group have prime order.

### Exercise 35.9: Composition series of $S_3 \times \mathbb{Z}_2$

Put $T = A_3 \times \{0\}$, $B = S_3 \times \{0\}$, $A = A_3 \times \mathbb{Z}_2$, $U = \{e\} \times \mathbb{Z}_2$, and $C = \{(\sigma, \epsilon) : \operatorname{sgn}(\sigma) = (-1)^\epsilon\}$. Then all four series are

$$1 < T < A < G, \qquad 1 < U < A < G,$$
$$1 < T < B < G, \qquad 1 < T < C < G.$$

Here $A \cong \mathbb{Z}_6$, while $B \cong C \cong S_3$. To check exhaustion, the commutator subgroup of $G$ is $T$, and $G/T \cong \mathbb{Z}_2^2$. A maximal normal subgroup must therefore be one of the three inverse images of its lines, namely $A, B, C$ (a simple quotient of this solvable group has prime order). The cyclic group $A$ has the two indicated series, and each $S_3$ factor has only its $A_3$ series. The factor orders are $3, 2, 2$ or $2, 3, 2$, so all are isomorphic.

### Exercise 35.10: Composition series of $\mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_7$

The product is cyclic of order 70 because 2, 5, and 7 are pairwise coprime. For each of the six permutations $(p, q, r)$ of $(2, 5, 7)$, take

$$0 < U_p < U_{pq} < G,$$

where $U_d$ is the unique subgroup of order $d$. In product notation $U_p$ is the coordinate subgroup for prime $p$, and $U_{pq}$ is the product of the $p$ and $q$ coordinate subgroups. These are all the series because simple cyclic factors have prime order; the successive factor orders must be a permutation of $2, 5, 7$.

### Exercise 35.11: The center of $S_3 \times \mathbb{Z}_4$

An element $(g, z)$ commutes with every $(h, w)$ if and only if $gh = hg$ for every $h$; the second coordinate already commutes because $\mathbb{Z}_4$ is abelian. Since $Z(S_3) = \{e\}$,

$$Z(S_3 \times \mathbb{Z}_4) = \{e\} \times \mathbb{Z}_4.$$

For example a transposition fails to commute with a different transposition, and a nonidentity 3-cycle fails to commute with a transposition, verifying the stated center of $S_3$.

### Exercise 35.12: The center of $S_3 \times D_4$

Write $D_4 = \langle r, s \rangle$ with $r^4 = s^2 = e$ and $sr = r^{-1}s$. Its center is $\{e, r^2\}$: $r$ and $r^3$ fail to commute with $s$, and every reflection fails to commute with $r$. Commutation in a product is coordinatewise, so

$$Z(S_3 \times D_4) = \{e\} \times \{e, r^2\}.$$

In the textbook's notation $e = \rho_0$ and $r^2 = \rho_2$.

### Exercise 35.13: Ascending central series of $S_3 \times \mathbb{Z}_4$

The first nontrivial term is $Z_1 = \{e\} \times \mathbb{Z}_4$. The quotient $G/Z_1 \cong S_3$ has trivial center. Taking the inverse image of that trivial center gives $Z_2 = Z_1$, and the same reasoning repeats. Thus the series, including its initial identity term, is

$$\{(e, 0)\} \leq \{e\} \times \mathbb{Z}_4 \leq \{e\} \times \mathbb{Z}_4 \leq \cdots .$$

It never reaches $G$. Repeated terms must be retained when describing the ascending central sequence.

### Exercise 35.14: Ascending central series of $S_3 \times D_4$

The first term is $Z_1 = \{e\} \times \{e, r^2\}$. Now $G/Z_1 \cong S_3 \times \mathbb{Z}_2^2$, whose center is $\{e\} \times \mathbb{Z}_2^2$. Its inverse image is $Z_2 = \{e\} \times D_4$. Since $G/Z_2 \cong S_3$ has trivial center, every later term equals $Z_2$. Thus

$$1 \leq \{e\} \times \{e, r^2\} \leq \{e\} \times D_4 \leq \{e\} \times D_4 \leq \cdots .$$

Each term is normal in $G$, as required for the construction.

### Exercise 35.15: Definition of a composition series

The proposed definition is correct, provided "maximal normal" means maximal among *proper* normal subgroups. The finite strict chain starts at $\{e\}$ and ends at $G$, and each $H_i \trianglelefteq H_{i+1}$ has no normal subgroup strictly between it and $H_{i+1}$. By the correspondence theorem this is equivalent to $H_{i+1}/H_i$ being simple.

### Exercise 35.16: Definition of solvability

Replace "a composition series of abelian groups" by "a composition series whose *factor groups* are abelian." The intermediate groups need not be abelian: $1 < A_3 < S_3$ is a solvable composition series even though $S_3$ is nonabelian. This uses the textbook's Definition 35.18. It restricts solvability to groups admitting such a finite composition series; the broader modern definition uses a finite subnormal series with abelian factors, allowing infinite groups such as $\mathbb{Z}$.

### Exercise 35.17: True or false: group series

**(a)** True. Normality in $G$ implies normality in every intermediate subgroup.

**(b)** False. In $D_4$, $1 < \langle s \rangle < \langle s, r^2 \rangle < D_4$ is subnormal, but $\langle s \rangle$ is not normal in $D_4$.

**(c)** True under Definition 35.12 in this text: a principal series is normal with simple factors, hence subnormal with simple factors.

**(d)** False. The composition series in (b) has order-2 factors but is not normal in $D_4$.

**(e)** False. $\mathbb{Z}_6$ has $0 < \langle 2 \rangle < \mathbb{Z}_6$ and $0 < \langle 3 \rangle < \mathbb{Z}_6$; moreover $\mathbb{Z}$ has no composition series.

**(f)** True. Repeatedly choose a maximal proper normal subgroup; orders strictly decrease, so the process terminates.

**(g)** False. Simple factors occur in every composition series by definition; $S_5$ has one but is not solvable, because $A_5$ is a nonabelian factor.

**(h)** False. $1 < A_7 < S_7$ has the nonabelian simple factor $A_7$.

**(i)** True. The simple factor groups, like primes in an integer factorization, are determined up to order and isomorphism, though the intermediate subgroups need not be unique.

**(j)** True. A group of prime order is cyclic, so its one factor is abelian and simple.

The textbook's definition of a principal/chief series here is stronger than the usual modern chief-series convention; the answers follow its stated convention.

### Exercise 35.18: A composition series of $S_3^2$

Take

$$1 < A_3 \times 1 < S_3 \times 1 < S_3 \times A_3 < S_3 \times S_3.$$

Each term is normal in the next: this follows from $A_3 \trianglelefteq S_3$ and coordinatewise conjugation. The factors, from bottom to top, are $\mathbb{Z}_3, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_2$. They are simple and abelian, so the series is a composition series and $S_3 \times S_3$ is solvable.

### Exercise 35.19: Solvability of the square-symmetry group

The chain $1 < \langle r^2 \rangle < \langle r \rangle < D_4$ has subgroup orders $1, 2, 4, 8$. The first subgroup is central, and the rotation subgroup has index 2, hence is normal. Every factor has order 2, so every factor is simple and abelian. Therefore $D_4$ is solvable. The definition does not require $D_4$ itself to commute.

### Exercise 35.20: Schreier arrays in $\mathbb{Z}_{36}$

**Plan.** Retain repeated entries while forming the rectangular Schreier arrays. In $\mathbb{Z}_n$, $\langle a \rangle + \langle b \rangle = \langle \gcd(a,b) \rangle$ and $\langle a \rangle \cap \langle b \rangle = \langle \operatorname{lcm}(a,b) \rangle$ when $a, b \mid n$; write $\langle n \rangle = 0$. Use $H_{ij} = H_i + (H_{i+1} \cap K_j)$ and $K_{ji} = K_j + (K_{j+1} \cap H_i)$. The arrays for chains (3) and (4) are respectively

$$\begin{array}{ccccc} 0 & \leq & 0 & \leq & \langle 12 \rangle \\ \langle 12 \rangle & \leq & \langle 6 \rangle & \leq & \langle 3 \rangle \\ \langle 3 \rangle & \leq & \langle 3 \rangle & \leq & G \end{array} \qquad \begin{array}{ccccccc} 0 & \leq & 0 & \leq & \langle 18 \rangle & \leq & \langle 18 \rangle \\ \langle 18 \rangle & \leq & \langle 6 \rangle & \leq & \langle 3 \rangle & \leq & G \end{array}.$$

Read each row left to right, identifying its last term with the first term of the next row. The factor-order array for the first chain is

$$\begin{pmatrix} 1 & 3 \\ 2 & 2 \\ 1 & 3 \end{pmatrix}.$$

The second chain has the transpose array of factor orders: the factor in row $i$, column $j$ of the first chain corresponds to row $j$, column $i$ of the second. All factors are cyclic; order 1 denotes a repeated group and thus a trivial factor. Deleting repeated entries gives the requested isomorphic refinements. This also exhibits every factor correspondence, rather than only their unordered lists.

### Exercise 35.21: Schreier arrays in $\mathbb{Z}_{24}$

**Plan.** Retain repeated entries while forming the rectangular Schreier arrays. In $\mathbb{Z}_n$, $\langle a \rangle + \langle b \rangle = \langle \gcd(a,b) \rangle$ and $\langle a \rangle \cap \langle b \rangle = \langle \operatorname{lcm}(a,b) \rangle$ when $a, b \mid n$; write $\langle n \rangle = 0$. Use $H_{ij} = H_i + (H_{i+1} \cap K_j)$ and $K_{ji} = K_j + (K_{j+1} \cap H_i)$. The arrays for chains (3) and (4) are respectively

$$\begin{array}{ccccccc} 0 & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle \\ \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 4 \rangle \\ \langle 4 \rangle & \leq & \langle 2 \rangle & \leq & G & \leq & G \end{array} \qquad \begin{array}{ccccccc} 0 & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 6 \rangle \\ \langle 6 \rangle & \leq & \langle 6 \rangle & \leq & \langle 6 \rangle & \leq & \langle 3 \rangle \\ \langle 3 \rangle & \leq & \langle 3 \rangle & \leq & G & \leq & G \end{array}.$$

Read each row left to right, identifying its last term with the first term of the next row. The factor-order array for the first chain is

$$\begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 3 \\ 2 & 2 & 1 \end{pmatrix}.$$

The second chain has the transpose array of factor orders: the factor in row $i$, column $j$ of the first chain corresponds to row $j$, column $i$ of the second. All factors are cyclic; order 1 denotes a repeated group and thus a trivial factor. Deleting repeated entries gives the requested isomorphic refinements. This also exhibits every factor correspondence, rather than only their unordered lists.

### Exercise 35.22: Normality in an intersection

Let $x \in H \cap K$ and $y \in H^* \cap K$. Since $x \in H$ and $H^* \trianglelefteq H$, the conjugate $xyx^{-1} \in H^*$. Since $x, y \in K$, it also lies in $K$. Thus conjugation by $x$ preserves $H^* \cap K$; using $x^{-1}$ yields equality. Therefore $H^* \cap K \trianglelefteq H \cap K$.

### Exercise 35.23: Multiplying finite factor orders

Start with $|H_0| = 1$. If $|H_i|$ is finite and $[H_{i+1} : H_i] = s_{i+1}$, then $H_{i+1}$ is the disjoint union of $s_{i+1}$ cosets, each with $|H_i|$ elements. Hence $|H_{i+1}| = s_{i+1}|H_i|$. Induction gives $|H_j| = s_1 \cdots s_j$ for every $j$. At $j = n$ this proves $|G| = s_1 \cdots s_n$. Normality is needed to call the factors groups, although the coset-counting argument itself only uses subgroup indices.

### Exercise 35.24: No infinite abelian composition series

A simple abelian group is cyclic of prime order: a nonidentity element generates a nontrivial normal subgroup, hence the whole group; an infinite cyclic group or a finite cyclic group of composite order has a proper nontrivial subgroup. If an abelian group had a composition series, all its factors would consequently be finite of prime order. Exercise 35.23 would make the group finite. This contradicts the hypothesis that it is infinite.

### Exercise 35.25: Finite products of solvable groups

For two factors choose composition series $1 = H_0 < \cdots < H_r = H$ and $1 = K_0 < \cdots < K_s = K$ with abelian factors. Form

$$1 < H_1 \times 1 < \cdots < H \times 1 < H \times K_1 < \cdots < H \times K.$$

Normality holds coordinatewise. The factors in the first portion are $H_i/H_{i-1}$; those in the second are $K_j/K_{j-1}$. Thus every factor is simple and abelian, so this is a solvable composition series. Apply the two-factor construction repeatedly for any finite number of groups.

### Exercise 35.26: Subgroups of solvable groups

Let $1 = H_0 < \cdots < H_n = G$ have abelian simple factors, hence cyclic factors of prime order. Set $K_i = K \cap H_i$. For $x \in K_i$, conjugation preserves $K \cap H_{i-1}$, so $K_{i-1} \trianglelefteq K_i$. The map $K_i \to H_i/H_{i-1}$, $x \mapsto xH_{i-1}$, has kernel $K_{i-1}$. Therefore $K_i/K_{i-1}$ is isomorphic to a subgroup of a prime-order cyclic group. It is either trivial or that entire prime-order group. After deleting repetitions the intersections form a composition series of $K$ with abelian factors. Thus $K$ is solvable. The prime-order property is essential: intersections with an arbitrary composition series need not have simple factors.

### Exercise 35.27: Inserting a simple normal subgroup

Because $N \trianglelefteq G$, each $H_iN$ is a subgroup. Also $H_{i-1}N \trianglelefteq H_iN$: conjugation by $H_i$ preserves both $H_{i-1}$ and $N$, while conjugation by $n \in N$ fixes every coset modulo $N$ and thus preserves $H_{i-1}N$. The composite map

$$H_i \longrightarrow H_iN/H_{i-1}N$$

is onto, with kernel $H_i \cap (H_{i-1}N)$ containing $H_{i-1}$. Hence its target is a quotient of the simple group $H_i/H_{i-1}$, and is either trivial or simple. Prepend 1 to $N = H_0N \leq H_1N \leq \cdots \leq G$. Its initial factor is the assumed simple group $N$; all later nontrivial factors are simple by the preceding argument. Delete repeated terms to obtain the required composition series.

### Exercise 35.28: Composition series descend to quotients

Let $\gamma : G \to G/N$ be the quotient map. For $x \in H_i$ and $y \in H_{i-1}$, $\gamma(x)\gamma(y)\gamma(x)^{-1} = \gamma(xyx^{-1}) \in \gamma(H_{i-1})$. Thus the image terms form a subnormal chain. The onto map

$$H_i/H_{i-1} \longrightarrow \gamma(H_i)/\gamma(H_{i-1}), \qquad hH_{i-1} \longmapsto \gamma(h)\gamma(H_{i-1})$$

is well defined, since $H_{i-1}$ maps into $\gamma(H_{i-1})$. A quotient of a simple group is either trivial or simple. The image chain starts at 1 and ends at $G/N$; deleting its repetitions therefore produces a composition series of $G/N$.

### Exercise 35.29: Homomorphic images preserve solvability

Let $f : G \to Q$ be a homomorphism, and use a composition series of $G$ with cyclic prime-order factors. Repeat the argument of Exercise 35.28 with the onto map $G \to f(G)$: each successive image factor is a quotient of a cyclic prime-order group, hence either trivial or cyclic of that prime order. Delete repetitions. The result is a composition series of $f(G)$ with abelian factors, proving that the image is solvable. No surjectivity onto the originally named codomain $Q$ is needed; the assertion concerns $f(G)$.